Lecture 24: Compound Simpson; Gauss and Gauss–Kronrod Rules

Math 6610

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Today's Goals

- Finish compound Simpson
- Gauss rules
- Gauss–Kronrod

Recall: Simpson's Rule on One Panel

On $[\alpha, \beta]$ with midpoint $\mu = \frac{\alpha+\beta}{2}$:

$$S(f) := \frac{\beta - \alpha}{6}\Big[f(\alpha) + 4f(\mu) + f(\beta)\Big].$$

Proposition. If $f \in C^4[\alpha, \beta]$, there is $\eta \in (\alpha, \beta)$ with

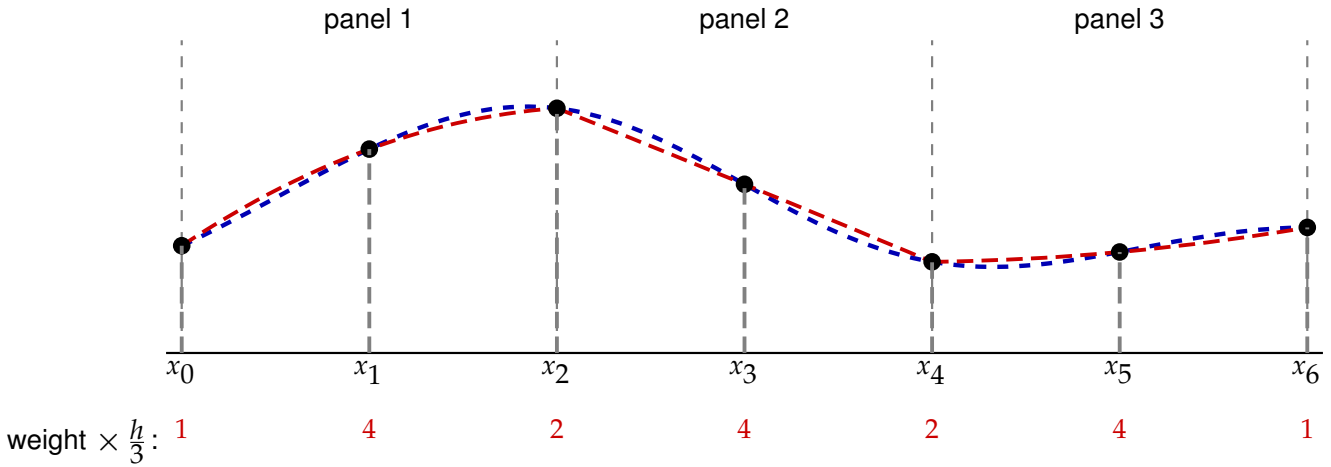
$$S(f) - \int_\alpha^\beta f(x)\,dx = \frac{(\beta - \alpha)^5}{2880} f^{(4)}(\eta).$$

- Exact for all cubics, using only 3 values of f.
- On a long interval the error is large, because of $(\beta - \alpha)^5$. Fix: cut $[a, b]$ into many short panels.

Compound Simpson: Setup

Definition (*Compound Simpson rule*). Let $s \geqslant 1$ be an integer, $h := \frac{b-a}{2s}$, $x_i := a + ih$ for $i = 0, 1, \ldots, 2s$. Simpson's rule on each of the s panels $[x_{2j-2}, x_{2j}]$ (width $2h$, midpoint x_{2j-1}) gives

$$S_{2s+1} := \frac{h}{3} \sum_{j=1}^{s} \left[f(x_{2j-2}) + 4f(x_{2j-1}) + f(x_{2j}) \right].$$



$s = 3$: blue f, red dashed the three parabolas, red numbers the weights.

Compound Simpson: The Weights

Each interior even point x_{2k}, $k = 1, \ldots, s-1$, is the right end of panel k and the left end of panel $k+1$: weight $1+1=2$. Hence

$$S_{2s+1} = \frac{h}{3}\left[f(a) + 4\sum_{j=1}^{s} f(x_{2j-1}) + 2\sum_{k=1}^{s-1} f(x_{2k}) + f(b)\right].$$

- Weights $\frac{h}{3}(1, 4, 2, 4, 2, \ldots, 2, 4, 1)$; for $s = 1$ the second sum is empty.
- Cost: $2s+1$ evaluations of f; the number of points is odd.
- Check: the weights sum to $\frac{h}{3}(2 + 4s + 2(s-1)) = 2sh = b - a$ (exact for $f \equiv 1$).

Compound Simpson in Matlab

```
function S = csimpson(f, a, b, s)
% compound Simpson with s panels, 2s+1 points
h = (b - a)/(2*s);
x = a + h*(0:2*s);
w = 2*ones(1, 2*s+1);          % even interior points
w(2:2:end) = 4;                % odd points x_1, x_3, ...
w([1 end]) = 1;                % end points
S = h/3 * sum(w .* f(x));
```

Usage, with $\int_0^\pi \sin x\,dx = 2$:

```
for s = 2.^(0:5)
  fprintf('%3d  %.3e\n', s, csimpson(@sin, 0, pi, s) - 2)
end
```

- `f` must accept a vector: write `@(x) x.^2`, not `@(x) x^2`.

Compound Simpson: Remainder

Let $I := \int_a^b f(x)\,dx$ and $R_{2s+1} := S_{2s+1} - I$.

Proposition. If $f \in C^4[a,b]$, there is $\xi \in (a,b)$ with

$$R_{2s+1} = (b-a)\frac{h^4}{180} f^{(4)}(\xi).$$

Proof. *1. One panel.* Panel j has width $\beta - \alpha = 2h$. By the one-panel remainder, for some $\xi_j \in (x_{2j-2}, x_{2j})$ the error is

$$\frac{(2h)^5}{2880} f^{(4)}(\xi_j) = \frac{h^5}{90} f^{(4)}(\xi_j).$$

2. Sum over the s panels.

$$R_{2s+1} = \frac{h^5}{90} \sum_{j=1}^{s} f^{(4)}(\xi_j) = \frac{s\,h^5}{90} \cdot \frac{1}{s} \sum_{j=1}^{s} f^{(4)}(\xi_j).$$

(continued on the next slide)

Compound Simpson: Remainder

Proof (continued). 3. One point ξ. The average $\frac{1}{s}\sum_j f^{(4)}(\xi_j)$ lies between the smallest and the largest of the values $f^{(4)}(\xi_j)$. Since $f^{(4)}$ is continuous, the intermediate value theorem gives ξ between two of the ξ_j (so $\xi \in (a,b)$) with $f^{(4)}(\xi)$ equal to the average.
4. Constants. Since $s = \frac{b-a}{2h}$,

$$\frac{s\,h^5}{90} = \frac{b-a}{2h} \cdot \frac{h^5}{90} = (b-a)\frac{h^4}{180}.$$

■

Remark. $|R_{2s+1}| \leqslant \frac{b-a}{180}\,h^4 \max_{[a,b]} \left|f^{(4)}\right|$. Halving h divides this bound by 16.

Convergence in Practice: The Ratio Test

$\int_0^\pi \sin x\,dx = 2$. Error E_N with N evaluations of f; ratio := E_N of the previous row divided by E_N of this row.

	compound trapezoid		compound Simpson	
N	error	ratio	error	ratio
3	$-4.29 \cdot 10^{-1}$		$9.44 \cdot 10^{-2}$	
5	$-1.04 \cdot 10^{-1}$	4.13	$4.56 \cdot 10^{-3}$	20.7
9	$-2.58 \cdot 10^{-2}$	4.03	$2.69 \cdot 10^{-4}$	16.9
17	$-6.43 \cdot 10^{-3}$	4.01	$1.66 \cdot 10^{-5}$	16.2
33	$-1.61 \cdot 10^{-3}$	4.00	$1.03 \cdot 10^{-6}$	16.1
65	$-4.02 \cdot 10^{-4}$	4.00	$6.45 \cdot 10^{-8}$	16.0

- Ratio $\to 4 = 2^2$: error $O(h^2)$. Ratio $\to 16 = 2^4$: error $O(h^4)$.
- Simpson's error is positive, as predicted: $f^{(4)} = \sin \geqslant 0$.
- *Debugging tip:* if your code does not show ratio ≈ 16 on a smooth f, the code has a bug.

When the Rate Fails: $f \notin C^4$

$\int_0^1 \sqrt{x}\,dx = \frac{2}{3}$. Here $f'(x) = \frac{1}{2\sqrt{x}}$ is unbounded at 0, so the proposition does not apply.

s	1	2	4	8	16	32
error (in 10^{-3})	-28.6	-10.1	-3.59	-1.27	-0.449	-0.159
ratio		2.82	2.83	2.83	2.83	2.83

- Ratio $2.83 \approx 2^{3/2}$: error $\approx Ch^{3/2}$, not h^4. Nearly all of it comes from the first panel $[0, 2h]$.
- Lesson: the rate is set by the smoothness of f, not by the rule. Uniform refinement wastes points where f is smooth.
- Fix: small panels only near $x = 0$ (adaptive quadrature, next lecture), or a change of variables ($x = t^2$ gives $\int_0^1 2t^2\,dt$, a polynomial).

Halving h: Error Estimate

In practice I is unknown. Compare S_{2s+1} (step h) with S_{4s+1} (step $\frac{h}{2}$). The remainder formula with $\frac{h}{2}$ gives, for some $\hat{\xi} \in (a, b)$,

$$R_{4s+1} = (b-a)\frac{(h/2)^4}{180} f^{(4)}(\hat{\xi}) = \frac{1}{16}(b-a)\frac{h^4}{180} f^{(4)}(\hat{\xi}).$$

Assumption (the only approximation): $f^{(4)}(\hat{\xi}) \approx f^{(4)}(\xi)$. Then $R_{4s+1} \approx \frac{1}{16} R_{2s+1}$. Since $S_{4s+1} - S_{2s+1} = R_{4s+1} - R_{2s+1}$,

$$S_{4s+1} - S_{2s+1} \approx -\tfrac{15}{16} R_{2s+1}, \qquad \text{so} \qquad |R_{2s+1}| \approx \tfrac{16}{15} |S_{4s+1} - S_{2s+1}| .$$

Remark. Accept $I \approx S_{4s+1}$ with error estimate $E := \frac{16}{15} |S_{4s+1} - S_{2s+1}|$. E estimates the error of the coarser S_{2s+1}, so it is about 16 times the error of S_{4s+1}: a conservative estimate.

Halving h: Reusing Values

Let $h = \frac{b-a}{2s}$.

Definition (*The sums A_{2s+1} and B_s*).

$$A_{2s+1} := f(a) + 2\sum_{i=1}^{2s-1} f(a+ih) + f(b) \qquad \text{(all points, interior ones doubled)}$$

$$B_s := 2\sum_{j=1}^{s} f\big(a+(2j-1)h\big) \qquad \text{(odd-indexed points)}$$

Proposition. $S_{2s+1} = \dfrac{h}{3}\big(A_{2s+1} + B_s\big)$.

Proof. In $A_{2s+1} + B_s$ the end points have weight 1, each even-indexed interior point has weight 2 (from A_{2s+1}), and each odd-indexed point has weight $2+2=4$. This is the pattern $(1,4,2,4,\ldots,4,1)$ of S_{2s+1}. ■

Halving h: Reusing Values

Step $\frac{h}{2} = \frac{b-a}{4s}$. The proposition with s replaced by $2s$ gives

$$S_{4s+1} = \frac{h/2}{3}\left(A_{4s+1} + B_{2s}\right),$$

where

$$A_{4s+1} = f(a) + 2\sum_{i=1}^{4s-1} f\left(a + i\tfrac{h}{2}\right) + f(b),$$

$$B_{2s} = 2\sum_{j=1}^{2s} f\left(a + (2j-1)\tfrac{h}{2}\right).$$

Proposition. $A_{4s+1} = A_{2s+1} + B_{2s}$.

Proof. Split the sum in A_{4s+1} by the parity of i. For $i = 2k$, $k = 1, \ldots, 2s-1$: $a + 2k\frac{h}{2} = a + kh$, so these terms form the sum in A_{2s+1}. For $i = 2j-1$, $j = 1, \ldots, 2s$: these terms form B_{2s}. ∎

Simpson from Two Trapezoid Sums

Let T_h be the compound trapezoid rule with step h on $[a, b]$.

Proposition. $T_h = \frac{h}{2} A_{2s+1}$, $T_{2h} = h\big(A_{2s+1} - B_s\big)$, and

$$S_{2s+1} = \frac{4T_h - T_{2h}}{3}.$$

Proof. 1. Weights. T_h has weights $\frac{h}{2}(1, 2, \ldots, 2, 1)$: this is $\frac{h}{2} A_{2s+1}$. $A_{2s+1} - B_s$ keeps only the end points (weight 1) and the even-indexed points (weight 2), i.e. the step-$2h$ grid; so $T_{2h} = \frac{2h}{2}(A_{2s+1} - B_s)$.
2. Combine.

$$\frac{4T_h - T_{2h}}{3} = \frac{2hA_{2s+1} - hA_{2s+1} + hB_s}{3} = \frac{h}{3}\big(A_{2s+1} + B_s\big) = S_{2s+1}.$$

■

Remark. Richardson extrapolation: it cancels the h^2 error term of the trapezoid rule. Repeating it gives Romberg integration.

Halving h: One Refinement Step

Remark (*Algorithm*). Given A_{2s+1}, S_{2s+1}, and a tolerance $\mathrm{tol} > 0$:

1. Compute B_{2s}: only $2s$ new evaluations of f (all other values are already in A_{2s+1}).
2. $A_{4s+1} = A_{2s+1} + B_{2s}$.
3. $S_{4s+1} = \frac{h/2}{3}\left(A_{4s+1} + B_{2s}\right)$.
4. $E = \frac{16}{15}\,|S_{4s+1} - S_{2s+1}|$. If $E \leqslant \mathrm{tol}$, accept S_{4s+1}; otherwise replace s by $2s$, h by $\frac{h}{2}$, and repeat.

Cost. After reaching s panels the total number of evaluations is $2s + 1$: exactly what S_{2s+1} alone costs. Every earlier estimate was free.

Refinement Loop in Matlab

```
function [S, E, s] = simpson_refine(f, a, b, tol, smax)
s = 1;  h = (b - a)/2;
A = f(a) + 2*f(a + h) + f(b);          % A_3
B = 2*f(a + h);                         % B_1
S = h/3*(A + B);                        % S_3
while s < smax
  h = h/2;                              % new step
  B = 2*sum(f(a + h*(1:2:4*s-1)));      % B_{2s}: 2s new values
  A = A + B;                            % A_{4s+1}
  Snew = h/3*(A + B);                   % S_{4s+1}
  E = 16/15*abs(Snew - S);
  S = Snew;  s = 2*s;
  if E <= tol, return, end
end
warning('tolerance not reached')
```

$\int_0^\pi \sin x\,dx$, tol $= 10^{-8}$: stops at $s = 128$ (257 values), $E = 4.0 \cdot 10^{-9}$, true error $2.5 \cdot 10^{-10}$.

Halving h: A Worked Check

Example. $f(x) = x^4$ on $[0,1]$, so $I = \frac{1}{5}$ and $f^{(4)} \equiv 24$.

$$
\begin{aligned}
s = 1,\ h = \tfrac{1}{2}: \quad & A_3 = 0 + 2 \cdot \tfrac{1}{16} + 1 = 1.125, \quad B_1 = 2 \cdot \tfrac{1}{16} = 0.125, \\
& S_3 = \tfrac{1}{6}(1.25) = 0.208333. \\
\tfrac{h}{2} = \tfrac{1}{4}: \quad & B_2 = 2\left(\tfrac{1}{256} + \tfrac{81}{256}\right) = 0.640625, \\
& A_5 = 1.125 + 0.640625 = 1.765625, \\
& S_5 = \tfrac{1}{12}(A_5 + B_2) = \tfrac{1}{12}(2.40625) = 0.200521.
\end{aligned}
$$

Check: $R_3 = 0.008333 = \frac{1}{120} = (1)\frac{(1/2)^4}{180} \cdot 24$, $R_5 = 0.000521 = \frac{R_3}{16}$, and $\frac{16}{15}|S_5 - S_3| = \frac{16}{15}(0.0078125) = 0.008333 = R_3$ (exact here because $f^{(4)}$ is constant).

Free Nodes: Degree of Exactness

Every rule so far has the form

$$I = \int_a^b f(x)\,dx \approx \sum_{i=1}^{n} w_i f(x_i). \tag{1}$$

Definition (*Degree of exactness*). Let $\prod_m$ be the polynomials of degree $\leqslant m$. Rule (1) has *degree of exactness* m if $\sum_i w_i p(x_i) = \int_a^b p(x)\,dx$ for all $p \in \prod_m$, but not for all $p \in \prod_{m+1}$.

- Newton–Cotes (trapezoid, Simpson): nodes fixed and equally spaced, only the n weights are free. Degree $n-1$ or n.
- Counting: n nodes $+$ n weights $= 2n$ parameters, and dim $\prod_{2n-1} = 2n$. This *suggests* degree $2n-1$, but proves nothing: the equations are nonlinear in the nodes.
- Gauss: pick the nodes cleverly, then the weights are linear.

Orthogonal Polynomials

Definition (*Inner product and orthogonal polynomial*). For polynomials p, q let $\langle p, q\rangle := \int_a^b p(x)q(x)\,dx$. The *$n$th orthogonal polynomial* is the monic (leading coefficient 1) $p_n(x) = x^n + \cdots \in \prod_n$ with

$$\langle q, p_n\rangle = 0 \qquad \text{for all } q \in \prod_{n-1}.$$

Construction: Gram–Schmidt on $1, x, x^2, \ldots,$

$$p_0 = 1, \qquad p_n = x^n - \sum_{j=0}^{n-1} \frac{\langle x^n, p_j\rangle}{\langle p_j, p_j\rangle}\, p_j.$$

On $[-1, 1]$ these are the monic Legendre polynomials:

$$p_1 = x, \qquad p_2 = x^2 - \tfrac{1}{3}, \qquad p_3 = x^3 - \tfrac{3}{5}x.$$

(Check: $\int_{-1}^{1}(x^2 - \frac{1}{3})\,dx = \frac{2}{3} - \frac{2}{3} = 0$.)

The Zeros of p_n

Lemma. For $n \geqslant 1$, p_n has n distinct real zeros, all in (a, b).

Proof. 1. Sign changes. Let $z_1, \ldots, z_m$ be the points of (a, b) where p_n changes sign. Then $m \leqslant n$, as $p_n \not\equiv 0$ has at most n zeros.
2. Assume $m \leqslant n-1$. Put $q(x) := \prod_{i=1}^{m}(x - z_i) \in \prod_{n-1}$ ($q \equiv 1$ if $m = 0$).
3. One sign. Each factor $x - z_i$ changes sign exactly where p_n does, so the continuous $p_n q \not\equiv 0$ has one sign on (a, b), and

$$\langle p_n, q \rangle = \int_a^b p_n(x) q(x)\, dx \neq 0.$$

4. Contradiction. Orthogonality gives $\langle p_n, q \rangle = 0$. Hence $m = n$: n distinct zeros in (a, b), and no others, since $\deg p_n = n$. ■

The Gauss Rule

Definition (*Gauss rule G_n*). Let $a < x_1 < \cdots < x_n < b$ be the zeros of p_n, and

$$w_i := \int_a^b \ell_i(x)\,dx, \qquad \ell_i(x) := \prod_{j \neq i} \frac{x - x_j}{x_i - x_j} \quad (i = 1, \ldots, n).$$

The *n-point Gauss rule* is $G_n(f) := \sum_{i=1}^n w_i f(x_i)$.

Remark (*Lecture 23*). These are the unique weights for which G_n is exact on $\prod_{n-1}$: $G_n(f) = \int_a^b f\,dx$ for all $f \in \prod_{n-1}$ (integrate the interpolant $\sum_i f(x_i)\ell_i = f$).

n	nodes on $[-1, 1]$	weights
1	0	2
2	± 0.577350 $(\pm 1/\sqrt{3})$	$1,\ 1$
3	$0,\ \pm 0.774597$ $(\pm\sqrt{3/5})$	$\frac{8}{9},\ \frac{5}{9},\ \frac{5}{9}$
4	$\pm 0.339981,\ \pm 0.861136$	$0.652145,\ 0.347855$
5	$0,\ \pm 0.538469,\ \pm 0.906180$	$0.568889,\ 0.478629,\ 0.236927$

Gauss Is Exact on $\prod_{2n-1}$

Theorem. $G_n(p) = \int_a^b p(x)\,dx$ for all $p \in \prod_{2n-1}$.

Proof. 1. Divide. Let $p \in \prod_{2n-1}$. Polynomial division by p_n (degree n) gives q, r with

$$p = p_n q + r, \qquad q, r \in \prod_{n-1}.$$

2. *Integrate.* $\int_a^b p_n q\,dx = \langle q, p_n \rangle = 0$, because $q \in \prod_{n-1}$. So $\int_a^b p\,dx = \int_a^b r\,dx$.
3. *Exact on r.* $r \in \prod_{n-1}$, so $\int_a^b r\,dx = \sum_i w_i r(x_i)$.
4. *Nodes are zeros.* $p_n(x_i) = 0$, so $p(x_i) = r(x_i)$ for each i. Together:

$$\int_a^b p(x)\,dx = \int_a^b r(x)\,dx = \sum_{i=1}^n w_i r(x_i) = \sum_{i=1}^n w_i p(x_i).$$

■

Degree $2n-1$ Is Optimal

Proposition. No rule (1) with n nodes is exact on $\prod_{2n}$. Hence G_n has degree of exactness exactly $2n-1$.

Proof. Let $x_1, \ldots, x_n$ be the nodes of any such rule and $q(x) := \prod_{i=1}^{n}(x-x_i)^2 \in \prod_{2n}$. Then $q \geqslant 0$ is continuous and $q \not\equiv 0$, so $\int_a^b q\,dx > 0$, while $\sum_i w_i q(x_i) = 0$. ∎

Example ($n=2$ on $[-1,1]$). $G_2(f) = f(-\frac{1}{\sqrt{3}}) + f(\frac{1}{\sqrt{3}})$. Exact for $1, x, x^2, x^3$; for x^4: $G_2(x^4) = \frac{1}{9} + \frac{1}{9} = \frac{2}{9} \neq \frac{2}{5} = \int_{-1}^{1} x^4 dx$.

n points	Newton–Cotes	Gauss
degree of exactness	$n-1$ (n even), n (n odd)	$2n-1$

Gauss Weights Are Positive

Proposition. $w_k = \int_a^b \ell_k(x)^2\,dx > 0$ for $k = 1, \ldots, n$.

Proof. *1. Exactness applies.* $\ell_k^2 \in \prod_{2n-2} \subset \prod_{2n-1}$, so by the theorem $G_n(\ell_k^2) = \int_a^b \ell_k^2\,dx$.
2. Evaluate the rule. $\ell_k(x_i) = 1$ if $i = k$, 0 otherwise, so $G_n(\ell_k^2) = \sum_i w_i \ell_k(x_i)^2 = w_k$.
3. Sign. $\ell_k^2 \geqslant 0$ is continuous and $\ell_k(x_k) = 1$, so $\int_a^b \ell_k^2\,dx > 0$. ■

Remark (*Why it matters: stability*). $f \equiv 1$ gives $\sum_i w_i = b - a$, so $\sum_i |w_i| = b - a$ for every n. If each $f(x_i)$ carries an error at most δ, G_n changes by at most $(b-a)\delta$. Newton–Cotes weights turn negative for $n \geqslant 9$ (and $\sum |w_i| \to \infty$): not used at high order.

Error of the Gauss Rule

Theorem (*proof omitted; Stoer–Bulirsch, §3.6*). If $f \in C^{2n}[a,b]$, there is $\xi \in (a,b)$ with

$$\int_a^b f(x)\,dx - G_n(f) = \frac{f^{(2n)}(\xi)}{(2n)!}\int_a^b p_n(x)^2\,dx.$$

Example (*Check: $n=2$, $f=x^4$ on $[-1,1]$*). Left: $\frac{2}{5}-\frac{2}{9}=\frac{8}{45}$. Right: $f^{(4)}\equiv 24$, so $\frac{24}{4!}\int_{-1}^{1}(x^2-\frac{1}{3})^2dx = \frac{2}{5}-\frac{4}{9}+\frac{2}{9}=\frac{8}{45}$.

- The proof interpolates f and f' at the n nodes (Hermite, degree $2n-1$) and uses the theorem on that interpolant.
- Needs $2n$ derivatives: for smooth f, the error drops faster than any power of n. For rough f, it does not (examples below).

Gauss on Any Interval $[a, b]$

Tables list nodes t_i and weights $\hat{w}_i$ for $[-1, 1]$ only.

Proposition. Let $\varphi(t) := \frac{a+b}{2} + \frac{b-a}{2} t$ (maps $[-1, 1]$ onto $[a, b]$), $x_i := \varphi(t_i)$, $w_i := \frac{b-a}{2} \hat{w}_i$. Then $\sum_i w_i p(x_i) = \int_a^b p(x)\, dx$ for all $p \in \prod_{2n-1}$.

Proof. 1. *Substitute* $x = \varphi(t)$, $dx = \frac{b-a}{2} dt$:

$$\int_a^b p(x)\, dx = \frac{b-a}{2} \int_{-1}^{1} p(\varphi(t))\, dt.$$

2. *Apply* G_n *on* $[-1, 1]$. φ is affine, so $p \circ \varphi \in \prod_{2n-1}$, and

$$\frac{b-a}{2} \int_{-1}^{1} p(\varphi(t))\, dt = \frac{b-a}{2} \sum_i \hat{w}_i\, p(\varphi(t_i)) = \sum_i w_i p(x_i).$$

■

Nodes and Weights in Matlab

Golub–Welsch: nodes $=$ eigenvalues of a symmetric tridiagonal matrix, weights from the first components of its eigenvectors.

```
function [t, w] = gauss(n)
% n-point Gauss-Legendre nodes t and weights w on [-1,1]
k = 1:n-1;
beta = k ./ sqrt(4*k.^2 - 1);          % recurrence coefficients
T = diag(beta, 1) + diag(beta, -1);    % Jacobi matrix
[V, D] = eig(T);
[t, i] = sort(diag(D));                % nodes
w = 2*V(1, i).^2;                      % weights (row)
```

Use on $[a, b]$:

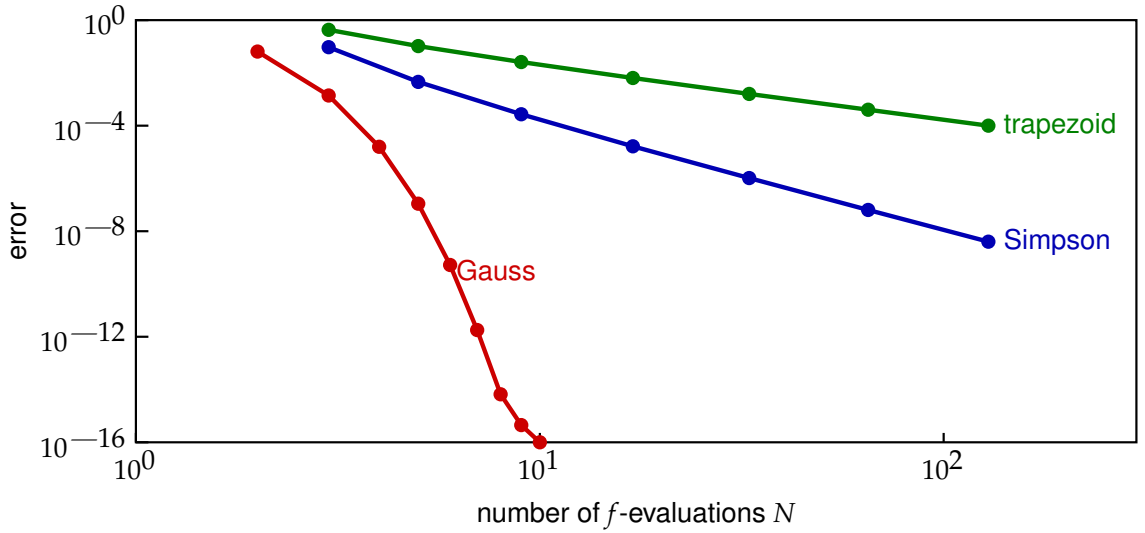
```
[t, w] = gauss(5);
x = (a+b)/2 + (b-a)/2*t;
Q = (b-a)/2 * w * f(x);
```

Why this works is an eigenvalue fact: proof in Lectures 25–29. Cost $O(n^2)$; $n = 5$ reproduces the table.

Gauss vs Simpson: Same Number of Evaluations

$\int_0^1 e^x\,dx = e - 1$, both rules with N evaluations of f:

N	compound Simpson	Gauss G_N
3	$5.8 \cdot 10^{-4}$	$8.2 \cdot 10^{-7}$
5	$3.7 \cdot 10^{-5}$	$6.5 \cdot 10^{-13}$
7	$7.3 \cdot 10^{-6}$	rounding level ($\leqslant 10^{-16}$)



$\int_0^\pi \sin x\,dx$, log–log: compound rules give lines (slopes -2, -4); Gauss falls faster than any line.

Gauss with an End-Point Singularity

$\int_0^1 \frac{dx}{\sqrt{x}} = 2$. Simpson cannot even start: $f(0)$ is infinite. Gauss can: all nodes lie in $(0, 1)$.

n	2	4	8	16	32	64
error	0.35	0.19	0.10	0.053	0.027	0.014

- The error only halves when n doubles: error $\approx C/n$. The error theorem does not apply ($f \notin C^1[0, 1]$).
- Still a correct answer, just slowly. Better: substitute $x = t^2$, $dx = 2t\,dt$: $\int_0^1 x^{-1/2} dx = \int_0^1 2\,dt$, which G_1 gets exactly.
- Rule of thumb: remove the singularity first (substitution), or let an adaptive code put small intervals near it.

Pros and Cons of Gaussian Quadrature

Pros	**Cons**
• Highest degree of exactness for n points: $2n-1$. • Positive weights: stable for every n. • No end-point values: works for f singular at a or b.	• When n is increased, the nodes move (G_2: ± 0.577; G_4: $\pm 0.340, \pm 0.861$): the old values $f(x_i)$ cannot be reused. • So an error estimate $\lvert G_{2n} - G_n \rvert$ costs $2n$ new evaluations.

Compare compound Simpson: halving h reuses every old value. Next: a Gauss type rule that reuses the Gauss values.

Gauss–Kronrod Rules: Definition

Keep the n Gauss nodes x_i (zeros of p_n); add $n+1$ new nodes.

Definition (*Stieltjes polynomial; Kronrod rule K_{2n+1}*). The *Stieltjes polynomial* is the monic $q_{n+1} \in \prod_{n+1}$ with

$$\int_a^b p_n(x)\, q_{n+1}(x)\, t(x)\, dx = 0 \qquad \text{for all } t \in \prod_n.$$

Its zeros $y_1, \ldots, y_{n+1}$ are the new nodes. The *Kronrod rule* is

$$K_{2n+1}(f) := \sum_{i=1}^{n} a_i f(x_i) + \sum_{j=1}^{n+1} b_j f(y_j),$$

with the weights a_i, b_j chosen so that K_{2n+1} is exact on $\prod_{2n}$.

Gauss–Kronrod Rules: Well Defined

Remark (q_{n+1} *exists and is unique*). Write $q_{n+1} = x^{n+1} + \sum_{j=0}^{n} c_j x^j$ and test with $t = x^k$, $k = 0, \ldots, n$: $n+1$ linear equations for $c_0, \ldots, c_n$ with matrix entries μ_{j+k}, $\mu_m := \int_a^b p_n x^m dx$. Orthogonality gives $\mu_m = 0$ for $m < n$, and $\mu_n = \langle p_n, p_n \rangle > 0$: the matrix is zero above its anti-diagonal, with $\mu_n \neq 0$ on it. So it is nonsingular.

Proposition (*Szegő, 1935; proof omitted*). For the interval $[a, b]$ (weight 1), the zeros y_j of q_{n+1} are real, simple, lie in (a, b), and interlace with the Gauss nodes: $y_1 < x_1 < y_2 < \cdots < x_n < y_{n+1}$.

Remark. So the $2n+1$ nodes are distinct, and the weights a_i, b_j exist and are unique (Vandermonde system, Lecture 23).

Kronrod Is Exact on $\prod_{3n+1}$

Theorem. $K_{2n+1}(p) = \int_a^b p(x)\,dx$ for all $p \in \prod_{3n+1}$.

Proof. *1. Divide* by $p_n q_{n+1}$ (degree $2n+1$): for $p \in \prod_{3n+1}$,

$$p = p_n q_{n+1} t + r, \qquad t \in \prod_n,\ r \in \prod_{2n}.$$

2. Integrate. $\int_a^b p_n q_{n+1} t\,dx = 0$ by the definition of q_{n+1}, since $t \in \prod_n$.
3. Exact on r. $r \in \prod_{2n}$, and the weights make K_{2n+1} exact on $\prod_{2n}$.
4. Nodes are zeros. $p_n(x_i) = 0$ and $q_{n+1}(y_j) = 0$, so $p = r$ at every node.
Together:

$$\begin{aligned}\int_a^b p\,dx = \int_a^b r\,dx &= \sum_{i=1}^{n} a_i r(x_i) + \sum_{j=1}^{n+1} b_j r(y_j) \\ &= \sum_{i=1}^{n} a_i p(x_i) + \sum_{j=1}^{n+1} b_j p(y_j).\end{aligned}$$

■

Why the Pair (G_n, K_{2n+1})

Parameter count. $2n+1$ weights $+\ n+1$ new nodes $= 3n+2 = \dim \prod_{3n+1}$: the theorem uses every free parameter.

Cost. G_n: n values. K_{2n+1}: the same n values plus $n+1$ new ones.

	new f-values	degree of exactness
G_n alone	n	$2n-1$
G_{2n} (to check G_n)	$2n$	$4n-1$
K_{2n+1} (to check G_n)	$n+1$	$3n+1$

Remark (*Practical use*). Report K_{2n+1} as the answer. Use $|G_n - K_{2n+1}|$ as error estimate: it estimates the error of the less accurate G_n, so it usually overestimates the error of K_{2n+1} (conservative). Standard pair in Matlab's `quadgk`: (G_7, K_{15}), 15 values per interval.

(G_7, K_{15}) on Examples

One interval, 15 evaluations of f each:

integral	error of G_7	error of K_{15}	$\lvert G_7 - K_{15}\rvert$
$\int_0^1 \cos 10x\, dx$	$1.2 \cdot 10^{-6}$	$1 \cdot 10^{-16}$	$1.2 \cdot 10^{-6}$
$\int_0^1 (1+25x^2)^{-1} dx$	$2.0 \cdot 10^{-5}$	$4.4 \cdot 10^{-9}$	$2.0 \cdot 10^{-5}$
$\int_0^1 \sqrt{x}\, dx$	$2.5 \cdot 10^{-4}$	$1.4 \cdot 10^{-5}$	$2.3 \cdot 10^{-4}$
$\int_{-1}^1 (1+25x^2)^{-1} dx$	$6.7 \cdot 10^{-2}$	$3.3 \cdot 10^{-3}$	$6.4 \cdot 10^{-2}$

- Each row: $|G_7 - K_{15}| \approx$ error of $G_7 >$ error of K_{15}. A safe estimate.
- Last row: one interval is not enough. Split $[a, b]$ where the estimate is large, repeat: adaptive quadrature (next lecture).
- QUADPACK sharpens it with an empirical formula: roughly $(200\,|G_7 - K_{15}|)^{3/2}$ when f is scaled to size about 1 (a heuristic, not a bound).

In Practice: Matlab's `integral` and `quadgk`

```
f = @(x) 1./(1 + 25*x.^2);          % vectorized!
q = integral(f, -1, 1)              % AbsTol 1e-10, RelTol 1e-6
q = integral(f, -1, 1, 'RelTol', 1e-12, 'AbsTol', 1e-14);
[q, errbnd] = quadgk(f, -1, 1)      % plus error estimate
q = integral(@(x) 1./sqrt(x), 0, 1)         % singular at 0
q = integral(@(x) exp(-x.^2), -Inf, Inf)    % infinite
```

- Both run (G_7, K_{15}) adaptively: split intervals where $|G_7 - K_{15}|$ is large.
- They never evaluate f at a or b; infinite intervals are mapped to finite ones by a change of variables first.
- Data instead of a formula (f known only at fixed points): `trapz(x, y)`, or integrate a spline: `integral(@(t) ppval(spline(x,y), t), a, b)`.

Summary: Rules and Their Exactness

rule	f-values	exact on	error
trapezoid	2	$\prod_1$	$\propto (b-a)^3 f''$
Simpson	3	$\prod_3$	$\propto (b-a)^5 f^{(4)}$
compound trapezoid	$m+1$	$\prod_1$ on each panel	$O(h^2), f \in C^2$
compound Simpson	$2s+1$	$\prod_3$ on each panel	$O(h^4), f \in C^4$
Gauss G_n	n	$\prod_{2n-1}$	$\propto f^{(2n)}$
Kronrod K_{2n+1}	$2n+1$	$\prod_{3n+1}$	–

- Smooth f: higher degree of exactness means faster convergence; Gauss gives the most accuracy per f-value.
- Rough f: all rates collapse (the $\sqrt{x}$ examples): adapt the grid to f.
- Error estimates come from comparing two rules that share f-values: (S_{2s+1}, S_{4s+1}) or (G_n, K_{2n+1}).

Looking Ahead

- **Next lecture:** Adaptive Quadrature; Eigenvalue Problems